**Idea Proposal: AI for Climate-Resilient Water Infrastructure**

**1. Project Idea:**

· **Problem Statement:** Climate change is exacerbating severe water scarcity in Syria, depleting groundwater and municipal supplies. Compounding this crisis is the degradation of aging urban water infrastructure, which suffers from high rates of non-revenue water (leaks and pipe bursts). Currently, municipalities rely on manual, reactive inspections, which are slow and result in the waste of millions of liters of already climate-stressed water.
· Specific Goal: This project aims to develop a Machine Learning anomaly detection model that identifies and localizes water leaks using time-series data from flow and pressure sensors. Because open-source, real-time data for Syrian networks is restricted, the model will be trained on a high-quality simulated/hybrid dataset. To bridge the "simulation-to-reality" gap, the project will utilize transfer learning techniques and validate the model against real-world benchmark datasets, ensuring the solution is robust and applicable to real-world Syrian utility scenarios.

**2. Relevance to Sustainable Development Goals (SDGs)**

· SDG 6 (Clean Water and Sanitation): Directly contributes to water conservation and efficient municipal water management.
· SDG 13 (Climate Action): Serves as a climate adaptation tool, protecting scarce water resources against the impacts of drought and climate change.
· SDG 9 (Industry, Innovation, and Infrastructure): Modernizes and upgrades water infrastructure using AI-driven predictive maintenance.
· SDG 11 (Sustainable Cities and Communities): Increases urban resilience by ensuring reliable access to essential utilities.

**3. Literature Examples**

· **Example 1:** "Machine Learning for Water Leakage Detection in Distribution Networks" (e.g., studies utilizing the BATADAL benchmark dataset). This research demonstrates how algorithms like Isolation Forests and Artificial Neural Networks (ANNs) can successfully identify pipe anomalies in time-series data. It will serve as the methodological blueprint for our model architecture and data preprocessing.
· **Example 2:** "Artificial Neural Network for Leak Detection in Syrian Water Networks" (Recent study from Tartous University, Syria). This regional study utilized real operational metrics (flow rate, pressure, vibration) at the Al-Jakara pumping station to predict leaks, achieving over 89% accuracy. This directly validates the feasibility of applying AI to water infrastructure in the Syrian context and supports our feature selection.